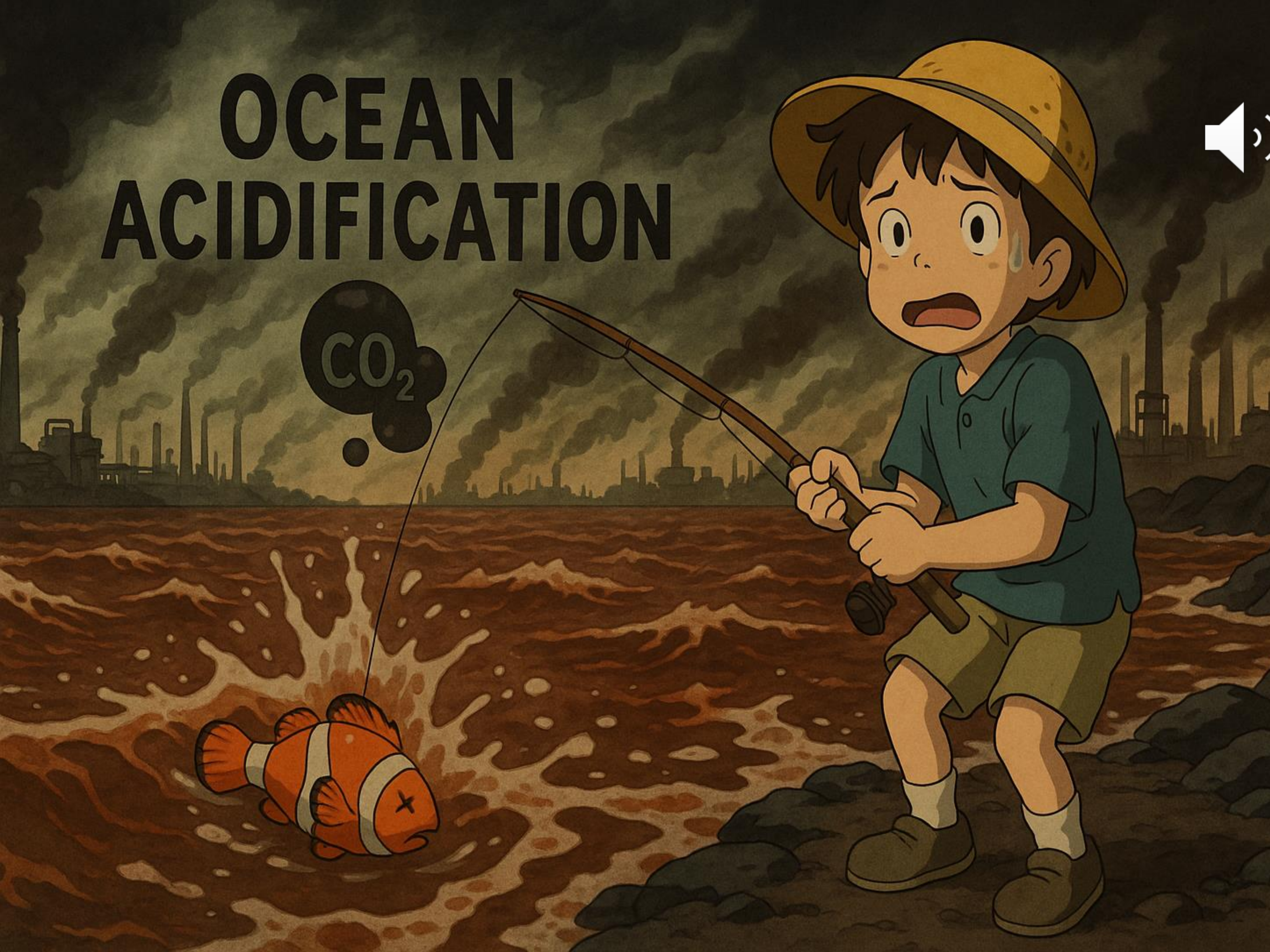
MY FIRST FISH & THE FUTURE OF THE OCEAN

By Caiden, Ryan & Tristan – 6/9/25

Story about catching my first fish to start and how that started my love for the ocean.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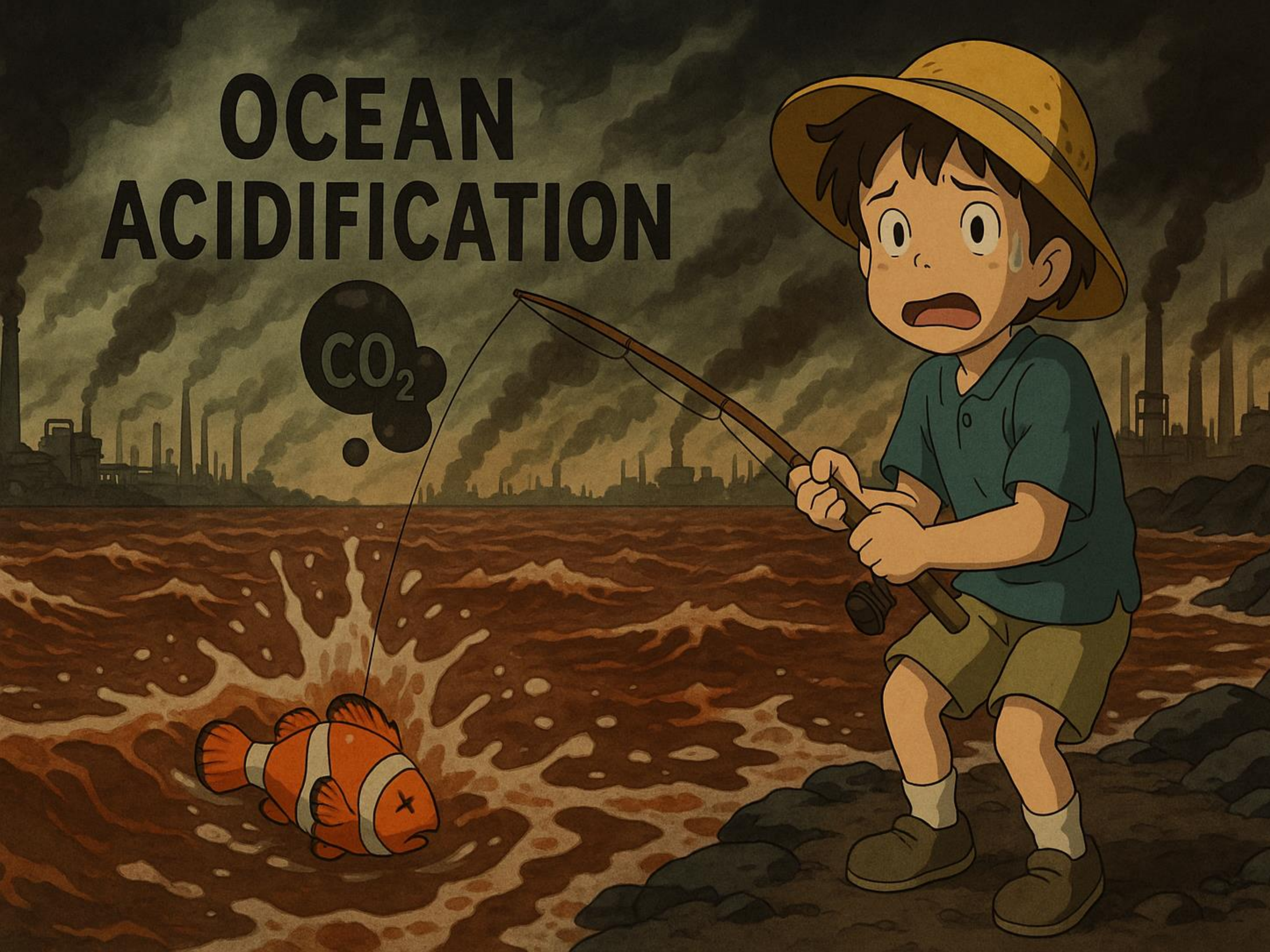
OCEAN
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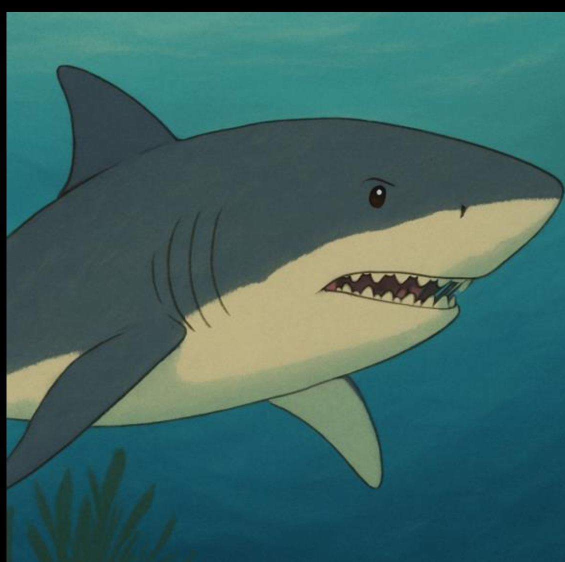
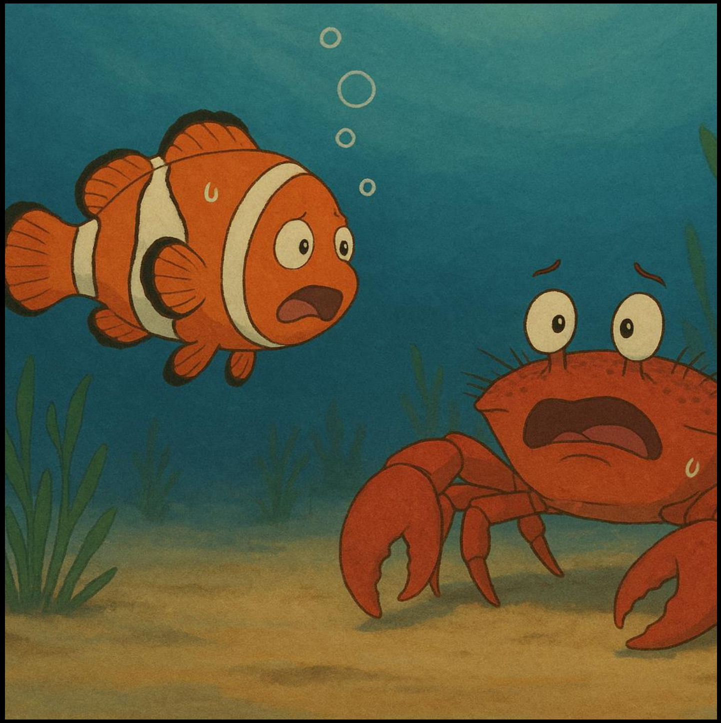
Take a clownfish, for example. These fish aren't just in Hollywood, they are also highly attuned to their environment. They rely on their sense of smell to detect predators, find food, and choose a safe habitat. However, in waters with a lower pH, their sense of smell is impaired. Studies show that acidified conditions confuse the chemical signals clownfish use to detect danger. Instead of swimming away from predators, now some clownfish swim right toward them.

It's not just clownfish being impacted. Dungeness crabs, a key species in the fishing industry and ecosystems, are also left vulnerable. They have tiny bristles, called mechanoreceptors, to sense their surroundings and find their way around the ocean floor. These bristles can detect changes in pressure and water movement, which helps the crab find food and avoid predators. These lower pH levels weaken these bristles and leave the crab unable to feel anything with them.

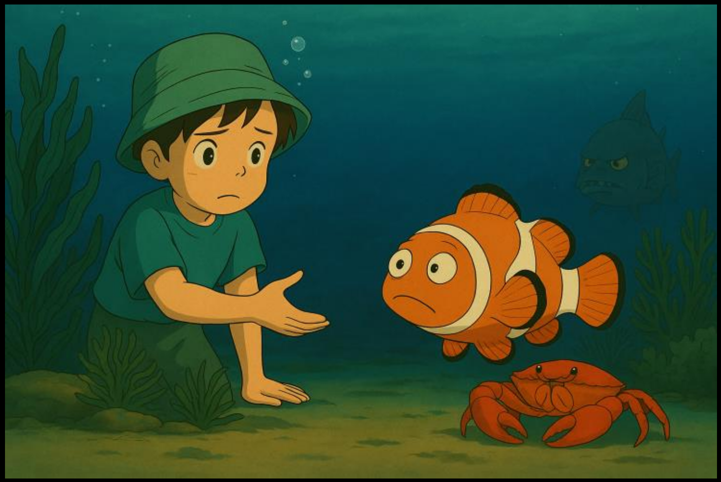




Clownfish and Dungeness crabs are very different, but their struggles under acidified waters show a shared challenge: trying to adapt to an ocean rapidly undergoing chemical change.

Researchers around the world are studying these effects to understand how acidification changes behavior, development, and survival rates in marine life. For clownfish, they look at the neurological pathways disrupted by acidic water. For Dungeness crabs, they are seeing how it affects shell formation and sensory organ development.

The implications of ocean acidification go much further than just individual species. Clownfish play a large role in maintaining coral reef health by eating small species that would otherwise overpopulate. Dungeness crabs are vital to food webs and support local fishing economies. Their decline could be the end of entire ecosystems.



But there is hope. Reducing CO_2 emissions through renewable energy and conservation efforts can help slow this down. Every step toward reducing our carbon footprint contributes to a healthier ocean.